Answer Key & Teacher Notes

Session 1 — Equivalent Ratios and Tables · Session 2 — The Bridge Strategy (Scaling Ratios)

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	About 10 minutes; use items 1–4 for a 5-minute check. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 · TEACHER NOTES

Equivalent Ratios and Tables

TALK PROMPTS

- How can you tell whether every column in a ratio table shows the same ratio?
- Why does adding the same number to both rows change the ratio, but multiplying does not?

PARTNER TASK

Lemonade Recipe. One partner extends the 3 : 2 table by multiplying; the other checks each column by dividing back to 3 : 2. Switch roles with the 4 : 3 pancake batter table.

WATCH FOR

- Adding the same number to both terms (writing 4 : 6 as 6 : 8) instead of multiplying.
- Multiplying only one row of the table by the scale factor.
- Finding the factor from the wrong row (dividing the new milk amount by the old mix amount).

SESSION 2 · TEACHER NOTES

The Bridge Strategy (Scaling Ratios)

TALK PROMPTS

- Why is this called the "Bridge Strategy"?
- Why can simplifying help you reach 20 blue tiles from 24 without a whole-number multiplier?

PARTNER TASK

Salad Dressing, Concrete Mix, Candle Scents. Partners take turns: one builds the bridge ratio and names the divisor; the other scales the bridge to the target and checks by dividing back to the bridge.

WATCH FOR

- Adding the same amount to both terms to reach the target (writing 4 : 6 as 10 : 12).
- Multiplying by a factor that only gets "close" to the target (12 × 2 = 24 instead of 20).
- Scaling only one term of the bridge ratio.

ITEM ANALYSIS

suggested DOK; judge the reasoning required, not the question label

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 · Form A	6.RP.A.3a	1	1	1	2	1	2	2	3	2	1	1
S1 · Form B	6.RP.A.3a	1	1	1	2	1	2	2	3	2	1	1
S2 · Form A	6.RP.A.3a	1	1	2	2	2	2	2	3	2	1	2
S2 · Form B	6.RP.A.3a	1	1	2	2	2	2	2	3	2	1	2

Items before the last three are Practice; then Find the Error (explain and correct) and two Test Practice items.

GUIDED NOTES

equivalent ratios: Ratios that make the same **comparison**.

ratio table: A table of **equivalent** ratios. Each column is one pair.

scale factor: The number you **multiply** or divide both terms by.

- To find an equivalent ratio, multiply or divide **both** terms by the **same** number.
- Adding the same number to both terms does **not** keep the ratio.

Example: Find the scale factor: 15 ÷ 5 = **3**. Multiply the water by the same factor: 2 × 3 = **6**. Check: 15 : 6 divided by 3 gives **5 : 2** again. Answer: 15 scoops of powder go with **6** cups of water.

YOU TRY

1. Water: 4, 8, 12, 20

2. a. 8 b. 3

EXIT TICKET

1. a. 15 b. 3

2. White: 4, 8, 12

3. **B** (2 : 3)

PRACTICE FORM A

1. a. 5 b. 6.4 c. 8 d. 15 e. 4 f. 5.5

2. Syrup: 2, 4, 6, 8; 18 lemons: 18 ÷ 3 = 6, 2 × 6 = 12 cups

3. Mix: 4, 8, 12, 20; Milk: 3, 6, 9, 15

4. Notebooks: 18, 9, 6, 3; Students: 6, 3, 2, 1

5. 45 ÷ 15 = 3, so 7 × 3 = 21 cups of sugar

6. 27 ÷ 4.5 = 6, so 9 × 6 = 54 pens

7. 4.5 ÷ 1.5 = 3, so 13.5 × 3 = 40.5 miles

8. a. Sample: 5 : 12 and 10 : 24 b. No. 4.5 : 8 adds 2 to each term. 2.5 × 1.8 = 4.5, but 6 × 1.8 = 10.8, not 8.

9. Adding 2 to both terms does not keep the ratio; you must multiply or divide both by the same number. 4 : 6 × 2 = 8 : 12 (or 4 : 6 ÷ 2 = 2 : 3).

10. **B** (8 : 20)

11. **C** (multiply both rows by the same number)

Self-check word **FACTOR**

PRACTICE FORM B

1. a. 7 b. 6.3 c. 8 d. 28 e. 5 f. 6.5

2. Syrup: 3, 6, 9, 12; 20 lemons: 20 ÷ 4 = 5, 3 × 5 = 15 cups

3. Oats: 5, 10, 15, 25; Honey: 2, 4, 6, 10

4. Books: 36, 12, 8, 4; Carts: 9, 3, 2, 1

5. 35 ÷ 7 = 5, so 4 × 5 = 20 cups of raisins

6. 21 ÷ 3.5 = 6, so 7 × 6 = 42 crayons

7. 7.5 ÷ 2.5 = 3, so 12.5 × 3 = 37.5 miles

8. a. Sample: 3 : 8 and 6 : 16 b. No. 3.5 : 6 adds 2 to each term. 1.5 : 4 = 3 : 8, but 3.5 : 6 = 7 : 12, which is a different ratio.

9. Adding 3 to both terms does not keep the ratio; you must multiply or divide both by the same number. 3 : 5 × 2 = 6 : 10.

10. **C** (9 : 12)

11. **B** (multiplied by 3)

Self-check word **SCALED**

GUIDED NOTES

bridge ratio: A **smaller** equivalent ratio you build first, then scale up.

simplify: Divide both terms by the **same** number.

greatest common factor: The **largest** number that divides both terms.

- When no whole number takes you to the target, **divide** both terms down to a bridge ratio.
- Then **multiply** both terms of the bridge up to the target.

Example: 6 × a whole number never equals **15**. Divide both terms by 2: 6 : 8 becomes **3 : 4**. Multiply both terms by 5: 3 : 4 becomes 15 : **20**. Answer: 15 cups of nuts go with **20** cups of raisins.

YOU TRY

1. 8 : 20 ÷ 4 = 2 : 5; × 7 = 14 : 35, so 35 spoons of water

2. a. 15 b. 12

EXIT TICKET

1. a. 3 : 5 b. 2 : 3

2. 6 : 4 ÷ 2 = 3 : 2; × 5 = 15 : 10, so 15 cups

3. **B** (divide both terms by the same number)

PRACTICE FORM A

1. a. 2 : 3 b. 3 : 4 c. 4 : 5 d. 4 : 9

2. Bridge 2 : 3 (÷ 2); 10 : 15 (× 5), so 15 cups of yellow

3. a. 6 : 9 ÷ 3 = 2 : 3 b. 2 : 3 × 7 = 14 : 21, so 21 tbsp

4. 9 : 15 ÷ 3 = 3 : 5; × 7 = 21 : 35, so 35 bags

5. 12 : 18 ÷ 6 = 2 : 3; × 5 = 10 : 15, so 15 drops

6. 24 : 30 ÷ 6 = 4 : 5; × 5 = 20 : 25, so 25 cream tiles

7. a. 12 ÷ 5 = 2.4 L b. 2.4 × 18 = 43.2 L

8. a. 10 ÷ 4 = 2.5 cups b. 2.5 × 18 = 45 cups c. 18 ÷ 4 = 4.5 is not a whole number, so bridge through 1 smoothie (or 5 : 2) instead.

9. Adding 6 to both terms does not keep the ratio. Bridge: 4 : 6 ÷ 2 = 2 : 3, then × 5 = 10 : 15. He needs 15 cups of yellow.

10. **D** (3)

11. **B** (15)

Self-check word **BRIDGES**

PRACTICE FORM B

1. a. 3 : 5 b. 2 : 3 c. 3 : 4 d. 4 : 9

2. Bridge 3 : 5 (÷ 2); 9 : 15 (× 3), so 15 cups of yellow

3. a. 8 : 12 ÷ 4 = 2 : 3 b. 2 : 3 × 10 = 20 : 30, so 30 cups

4. 21 : 28 ÷ 7 = 3 : 4; × 5 = 15 : 20, so 20 oz

5. 10 : 15 ÷ 5 = 2 : 3; × 8 = 16 : 24, so 24 white beads

6. 16 : 36 ÷ 4 = 4 : 9; × 3 = 12 : 27, so 27 cups

7. a. 15 ÷ 4 = 3.75 L b. 3.75 × 10 = 37.5 L

8. a. 15 ÷ 6 = 2.5 cups b. 2.5 × 14 = 35 cups c. 14 ÷ 6 is not a whole number, so bridge through 1 smoothie (or 5 : 2) instead.

9. Adding 2 to both terms does not keep the ratio. Bridge: 8 : 12 ÷ 4 = 2 : 3, then × 5 = 10 : 15. She needs 15 cups of white.

10. **B** (4.5)

11. **C** (25)

Self-check word **SCALING**

SHEET 1 · EQUIVALENT RATIOS AND TABLES

1. Nuts: 2, 4, 6, 10, 14; Cereal: 5, 10, 15, 25, 35
2. a. 20 b. 36 c. 4 d. 14
3. A, B, D (4 : 6 ÷ 2 = 2 : 3; × 2 = 8 : 12; 2 : 3 × 5 = 10 : 15; 6 : 8 adds 2 to each term)
4. 9 × 4 = 36, so 4 × 4 = 16 gold beads (table: 4 : 9, 8 : 18, 12 : 27, 16 : 36)
5. a. 3 × 7 = 21, so 2 × 7 = 14 cups of white b. 21 + 14 = 35 cups
6. No. Adding 3 to both terms does not keep the ratio; you must multiply or divide both terms by the same number. Example: 5 : 8 = 10 : 16 (× 2).

SHEET 2 · THE BRIDGE STRATEGY (SCALING RATIOS)

1. 2 minutes: 5 bottles (÷ 3); 10 minutes: 25 bottles (× 5)
2. a. 15 b. 8
3. a. 4 : 30 ÷ 2 = 2 : 15 b. 2 : 15 × 5 = 10 : 75, so 75 km
4. a. 10 : 4 ÷ 2 = 5 : 2 b. 5 : 2 × 3 = 15 : 6, so 15 cups
5. Adding 9 to both terms does not keep the ratio. Bridge: 6 : 10 ÷ 2 = 3 : 5, then × 5 = 15 : 25. She needs 25 spoons of water.
6. **A** (10)
7. **A** (÷ 3, then × 5)

October 2026 conceptual additions — teacher key

Session 1 · Equivalent Ratios and Tables

Seeds: 4, 6, 8. Multiply both terms by 2, 3, or 4. Adding 3 gives 6:5, while the equivalent ratio for 6 cups oats is 6:4.

Extension: Example: 1.5 cups oats and 1 cup seeds; divide both terms by 2.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.

Session 2 · The Bridge Strategy (Scaling Ratios)

Bridge: 6:8 ÷ 2 = 3:4, then × 5 = 15:20. Direct factor 15 ÷ 6 = 2.5; 8 × 2.5 = 20. Both preserve the ratio.

Extension: Divide 6:8 by 4 to get 1.5:2; 2 cups raisins. Efficiency judgments may differ if the arithmetic is valid.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.